small black holes are incredibly hot. The formula for the temperature of a black hole's radiation is engraved on Hawking's gravestone in West-minster Abbey.

Particles radiated by a black hole carry away energy, causing the hole to lose mass and eventually evaporate away completely. While it would be nice to observe Hawking radiation in a telescope, it's not going to happen for any of the black holes we know about. The Hawking temperature of a black hole the mass of the sun would be about 0.00000006 Kelvin. Any such signal would be swamped by other sources, such as the leftover microwave radiation from the Big Bang, which has a temperature of about 2.7 Kelvin. Even if such a black hole never grew by accreting matter and radiation, it will take over 10^{67} years for it to evaporate away completely.

There is a standard story that is told to explain why black holes emit radiation. I've told it, Hawking has told it, everyone tells it. It goes like this: according to quantum field theory, the vacuum is a bubbling stew of particles popping in and out of existence, typically in pairs consisting of one particle and one anti-particle. Ordinarily we don't notice, but in the vicinity of a black hole event horizon, one of the particles can fall inside the hole and then never get out, while the other escapes to the outside world. From the perspective of someone watching from afar, the escaping particle has positive energy, so to balance the books the infalling particle must have negative energy, and the black hole shrinks in mass as it absorbs these negative-energy particles.

Given our wave-functions-first Everettian perspective, there's a more accurate way to describe what's happening. The particles-appearing-and-disappearing story is a colorful metaphor that often provides physical intuition, and this is definitely one of those cases. But what we really have is a quantum wave function of the fields near the black hole. And that wave function is not static; it evolves into something else, in this case a smaller black hole plus some particles traveling away from it in all directions. It's not that different from an atom whose electrons have a bit of extra energy, and which therefore drop down to lower-energy states by emitting photons. The difference is that the atom eventually reaches a state of lowest possible energy and stays there, while the black hole (as far as we understand) just decays away entirely, exploding at the last second in a flash of high-energy particles.